

NAGAGAMI, ON, Canada

WMO: 718320

Lat: 49.7467N

Lon: 84.1636W

Elev: 866

StdP: 14.24

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.1	-25.6	-38.2	0.6	-29.9	-32.4	0.9	-25.2	16.9	13.7	15.5	12.5	3.8	300	0.646

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.1	84.2	67.3	80.9	65.1	77.9	63.9	70.6	79.6	68.4	76.6	66.4	73.9	10.3	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.7	105.3	74.5	65.4	97.1	72.5	63.4	90.1	70.9	35.1	80.1	33.3	76.3	31.6	74.1	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
16.4	14.3	12.8	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-38.8	89.5	5.9	2.8	-43.0	91.5	-46.4	93.2	-49.7	94.8	-54.0	96.9		
			-38.2	73.5	4.7	1.6	-41.6	74.7	-44.4	75.7	-47.0	76.6	-50.5	77.8		
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Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	34.2	-0.1	4.2	16.5	31.8	47.5	58.4	63.3	60.8	53.0	39.5	24.3	9.8
	DBStd	24.56	14.75	13.28	14.42	10.62	9.74	8.19	6.68	6.91	8.62	8.85	12.03	14.23
	HDD50	7048	1552	1284	1041	548	168	19	1	4	65	348	772	1246
	HDD65	11380	2017	1704	1505	995	551	229	115	170	371	791	1221	1711
	CDD50	1292	0	0	1	3	90	270	412	338	154	23	1	0
	CDD65	147	0	0	0	0	7	30	61	39	10	0	0	0
	CDH74	1649	0	0	0	2	136	434	610	375	90	2	0	0
	CDH80	375	0	0	0	0	34	107	148	74	12	0	0	0
Wind	WSAvg	6.2	5.8	6.4	6.5	6.4	6.5	5.9	5.9	5.9	6.1	6.5	6.3	5.8
Precipitation	PrecAvg	34.3	2.3	1.7	1.7	2.6	3.0	3.3	3.5	3.3	4.0	3.5	2.8	2.4
	PrecMax	40.9	4.4	3.7	3.0	4.6	4.8	5.4	6.2	6.9	6.1	6.3	4.6	4.0
	PrecMin	25.5	1.2	0.6	0.4	1.2	1.6	1.5	1.2	1.6	1.9	1.6	1.5	1.3
	PrecStd	3.9	0.7	0.7	0.6	1.0	0.8	0.9	1.2	1.3	1.2	1.3	0.8	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.7	40.3	57.5	68.4	84.6	88.2	88.7	86.0	81.7	71.5	55.2	39.8
		MCWB	34.2	35.6	47.9	50.5	64.3	67.8	70.2	69.3	66.5	59.7	49.7	37.3
	2%	DB	32.7	34.9	49.0	60.8	77.5	82.8	84.0	81.9	76.2	64.7	48.9	36.2
		MCWB	31.8	32.1	41.0	46.1	58.7	64.5	68.1	67.2	64.2	56.1	44.5	34.5
	5%	DB	29.5	30.6	43.0	54.9	71.9	79.2	80.4	78.3	71.9	59.4	44.1	33.2
		MCWB	28.4	28.4	36.7	43.2	55.9	62.6	65.9	65.9	62.9	53.1	40.6	32.1
	10%	DB	24.2	26.2	37.6	48.9	66.5	75.0	77.1	74.7	67.7	54.1	39.8	30.3
		MCWB	23.2	24.5	32.8	39.7	53.0	60.8	64.6	64.0	60.6	49.8	37.6	29.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.6	36.5	51.4	53.4	67.4	72.5	73.8	72.7	69.4	61.4	51.3	38.0
		MCDB	35.4	39.2	55.6	65.2	79.7	83.4	82.8	80.8	76.6	68.5	54.6	39.2
	2%	WB	32.0	32.4	41.9	47.4	61.7	68.4	70.9	69.8	66.6	57.6	45.2	34.5
		MCDB	32.7	34.5	47.8	58.6	72.9	77.2	80.2	78.4	72.6	63.4	47.8	35.6
	5%	WB	28.6	28.4	36.8	43.9	58.1	65.6	68.4	67.6	64.2	53.8	40.9	32.1
		MCDB	29.6	30.4	41.8	52.6	68.3	73.6	76.5	75.4	70.0	59.0	43.4	33.0
	10%	WB	23.3	24.5	33.2	40.4	54.7	63.0	66.2	65.3	61.1	49.6	37.6	29.4
		MCDB	24.3	26.3	37.4	48.3	64.6	71.9	74.0	72.9	67.6	53.8	39.7	30.2
Mean Daily Temperature Range		MDBR	20.2	23.6	25.3	22.6	24.0	23.6	22.1	21.6	19.5	14.2	13.6	16.5
		MCDBR	16.5	23.3	25.3	32.0	33.5	29.4	25.9	25.8	24.8	23.9	17.5	14.9
		MCWBR	15.9	20.4	18.0	19.2	17.8	14.6	13.0	13.7	14.7	16.2	13.6	14.1
	5% WB	MCDDBR	16.3	21.4	21.7	28.3	27.3	23.1	21.8	22.1	21.5	22.1	16.2	14.7
		MCWBR	15.9	19.2	15.9	18.2	15.8	13.6	12.7	13.1	14.4	16.2	13.3	14.2
Clear-Sky Solar Irradiance	taub	0.269	0.261	0.293	0.346	0.360	0.364	0.391	0.369	0.354	0.311	0.277	0.265	
	taud	2.245	2.311	2.294	2.319	2.402	2.414	2.359	2.429	2.463	2.557	2.507	2.310	
	Ebn at Noon	246	284	291	285	285	284	273	275	268	264	249	232	
	Edh at Noon	23	27	33	37	36	36	37	33	29	22	18	19	
All-Sky Solar Radiation	RadAvg	389	741	1171	1522	1637	1787	1743	1456	1042	612	328	262	
	RadStd	53	58	83	113	128	148	148	108	91	83	30	41	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page